Lecture 3

Axiomatic Definition of the Probability

23.02.2026

Outline

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Axiomatic probability

Definition

A probability space consists of a sample space S and a probability function P which takes an event $A \subset S$ as input and returns$P(A)$, a real number between 0 and 1, as output. The function P must satisfy the following axioms:

1. *For any event A, the probability of $0 \leq P(A) \leq 1$.*
2. $P(S) = 1$.
3. *If $A_1, A_2, \ldots$ are disjoint events, then*

$$P\left(\bigcup_{j=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

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Propositions

Complement Rule:

$$P(A^c) = 1 - P(A)$$

Proof (hint: use Axiom 2 and Axiom 3)

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- $P(\emptyset) = 0$

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- $P(B) \geq P(A)$, if $A \subseteq B$

Proof (hint: use Axiom 1)

Inclusion-Exclusion: two events A, B (not necessarily disjoint)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof (hint: use Axiom 3 and Venn-diagram)

Inclusion-Exclusion Example

J is taking two books along on her holiday vacation. With probability .5, she will like the first book; with probability .4, she will like the second book; and with probability .3, she will like both books. What is the probability that she likes neither book?

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1. Conditions: event $B_1 = \{\text{J likes book 1}\}$, $B_2 = \{\text{J likes book 2}\}$,

$$P(B_1) = 0.5, P(B_2) = 0.4, P(B_1 \cap B_2) = 0.3$$

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2. Question: find $P(B_1^c \cap B_2^c) = 1 - P(B_1 \cup B_2)$.
3. Formula: inclusion-exclusion

$$P(B_1 \cup B_2) = P(B_1) + P(B_2) - P(B_1 \cap B_2) = 0.5 + 0.4 - 0.3 = 0.6$$

$$1 - P(B_1 \cup B_2) = 1 - 0.6 = 0.4$$

Propositions

- Inclusion-Exclusion: three events A, B and c (not necessarily disjoint)

$$\begin{aligned} P(A \cup B \cup C) = & P(A) + P(B) + P(C) - P(A \cap B) \\ & - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C) \end{aligned}$$

Propositions

- Inclusion-Exclusion: n events $A_1, A_2, \dots A_n$ (not necessarily disjoint)

$$
\begin{aligned}
P(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i) - \sum_{1\leq i\leq j\leq n} P(A_i \cap A_j) \\
+ \sum_{1\leq i\leq j\leq k\leq n} P(A_i \cap A_j \cap A_k) + \cdots + \\
(-1)^{(n-1)} P(A_1 \cap A_2 \cap \dots A_n)
\end{aligned}
$$

Example

Pick an integer in $[1, 1000]$ at random. Compute the probability that it is divisible neither by 12 nor by 15.

Example

A large company with n employees has a scheme according to which each employee buys a Christmas gift and the gifts are then distributed at random to the employees. What is the probability that someone gets his or her own gift?

Probability: Frequentist interpretation

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- There is no agreement, at all, in how probabilities should be interpreted.
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- Frequentist interpretation: The probability of event A is the proportion of times (frequency) that A occurs in an infinite sequence (or very long run) of separate tries of the experiment.

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- Often associated with Jerzy Neyman and Egon Pearson who described the logic of statistical hypothesis testing.

Probability: Bayesian interpretation

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- Named after the 18^{th} century Presbyterian minister and mathematician Thomas Bayes.
- Largely popularized by revolutionary advance in computational technology and methods during the last twenty years.

Interpretations of Probability

- The Bayesian and frequentist perspectives are complementary, and both will be helpful for developing intuition in later chapters.
- Regardless of how we choose to interpret probability, we can use the two axioms to derive other properties of probability, and these results will hold for any valid probability function.

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Events A and B are independent if $P(A \cap B) = P(A)P(B)$.

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- If A and B are independent, then learning that B occurred gives us no information that would change our probabilities for A occurring
- Independence is completely different from disjointness. If A and B are disjoint, then $P(A \cap B) = 0$, so disjoint events can be independent only if $P(A) = 0$ or $P(B) = 0$. Knowing that A occurs tells us that B definitely did not occur, so A clearly conveys information about B, meaning the two events are not independent (except if A or B already has zero probability).

Proposition

If A and B are independent events, then so are A^c and B, A and B^c, A^c and B^c are also indepent.

Proof:

Example

An Internet packet travels from its source to router 1, from router 1 to router 2, and from router 2 to its destination. If routers drop packets independently with probability p, what is the probability that a packet is successfully transmitted from its source to its destination?